

Gebze Technical University

Dep. of Mechanical Engineering

Statics (ME 231)

Lecture 10:

Frames and Machines

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Frames and Machines

Frames and **machines** are two types of structures that are often composed of pin-connected ***multi-force members***, i.e., members that are subjected to more than two forces.

Frames are used to support loads:



This crane is a typical example of a framework.

Frames and Machines

Machines contain moving parts and are designed to transmit and alter the effect of forces.

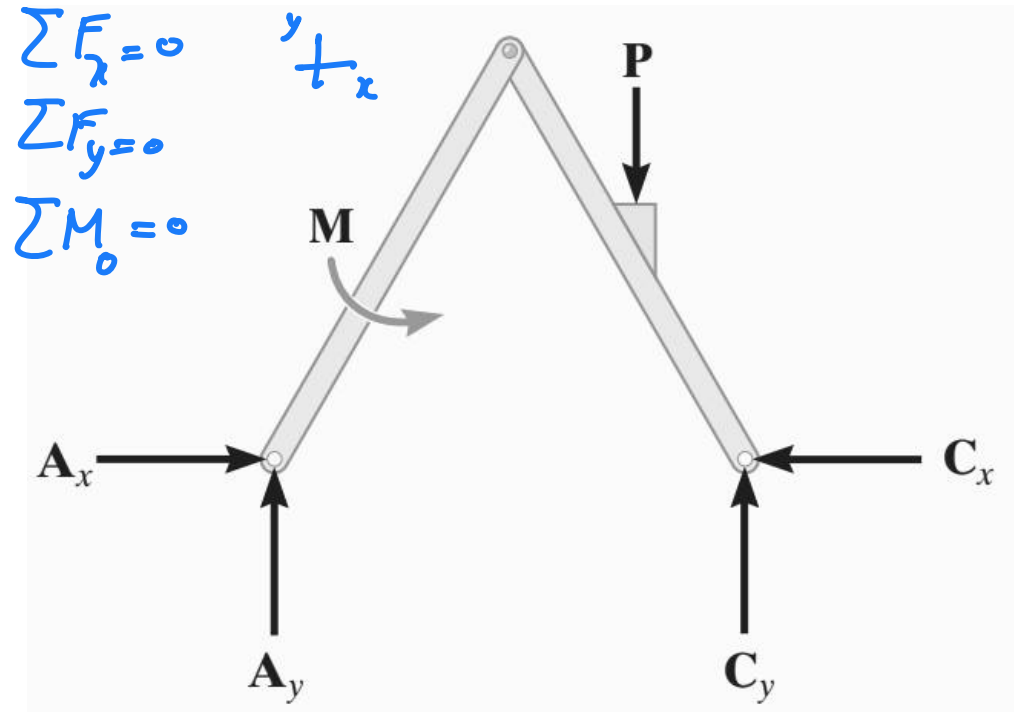
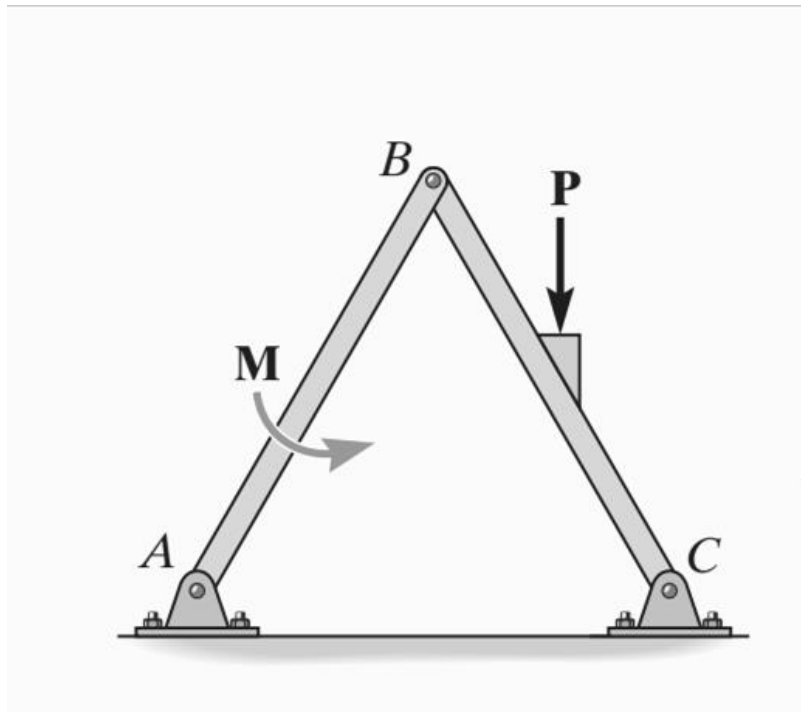


Common tools such as these pliers act as simple machines. Here the applied force on the handles creates a much larger force at the jaws.

Frames and Machines: Free-Body Diagrams

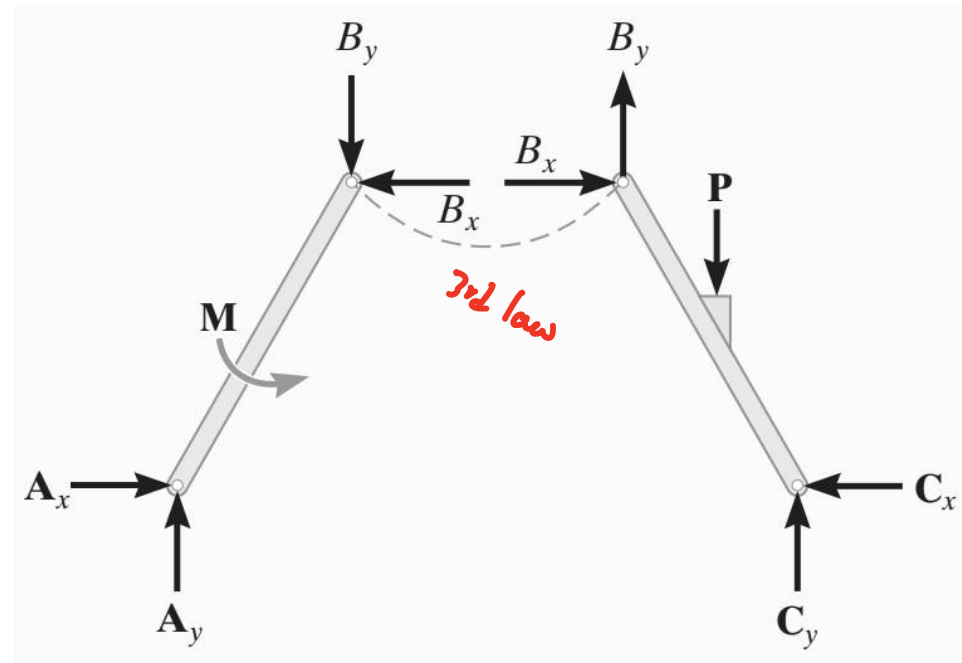
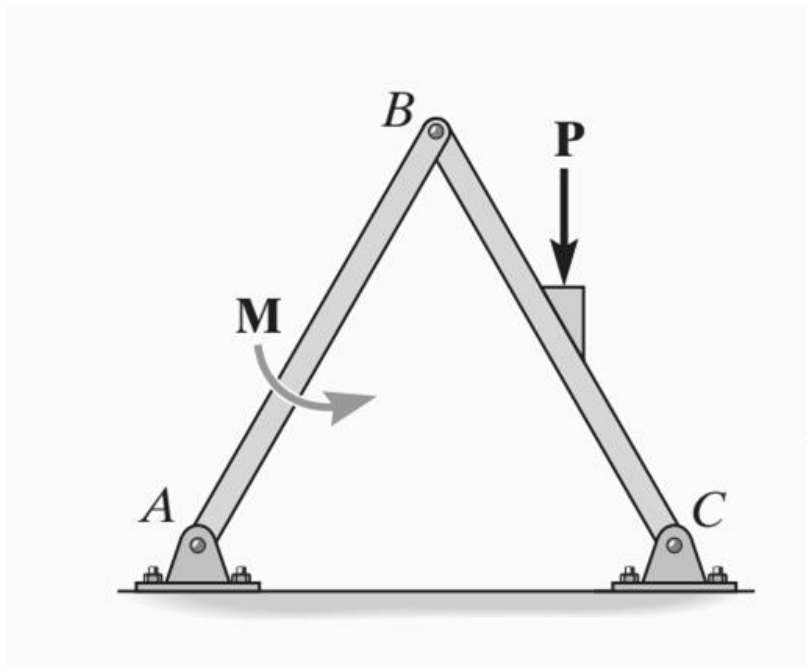
In order to determine the forces acting at the joints and supports of a frame or machine, the structure must be disassembled, and the free-body diagrams of its parts must be drawn.

Frame example:



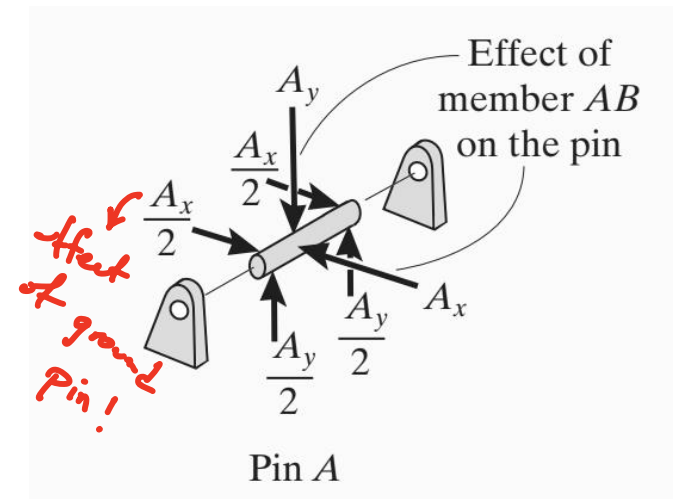
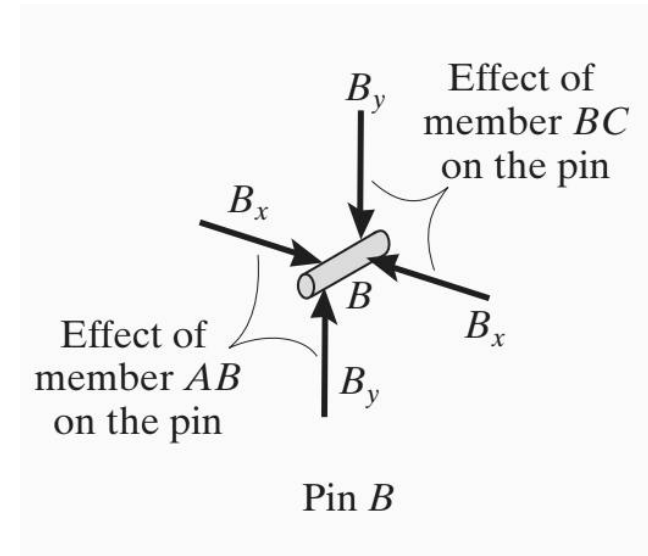
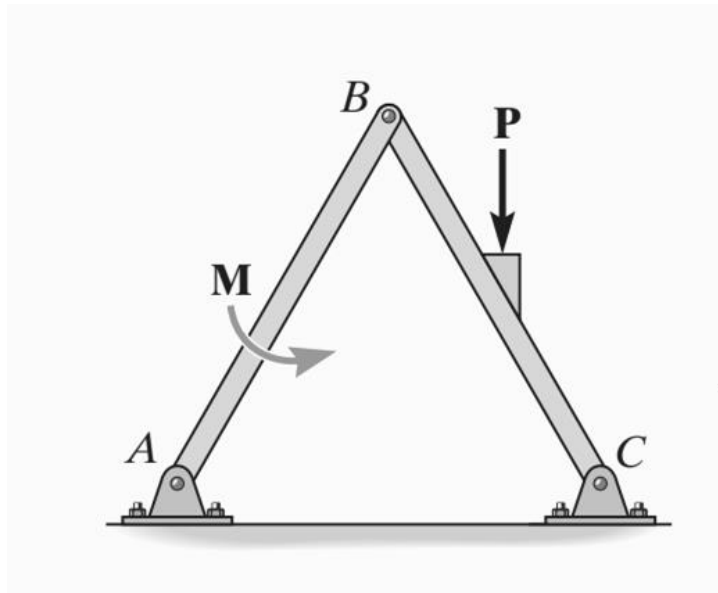
Frames and Machines: Free-Body Diagrams

Frame example:



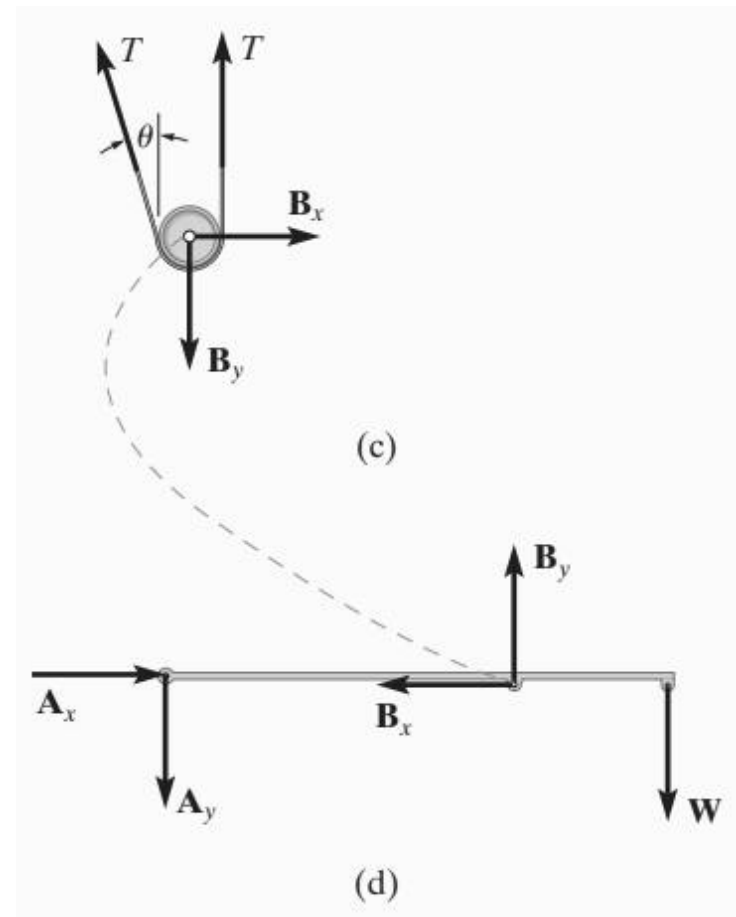
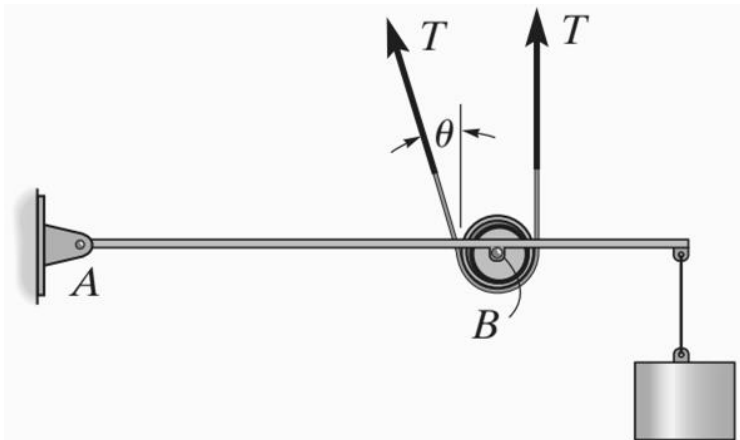
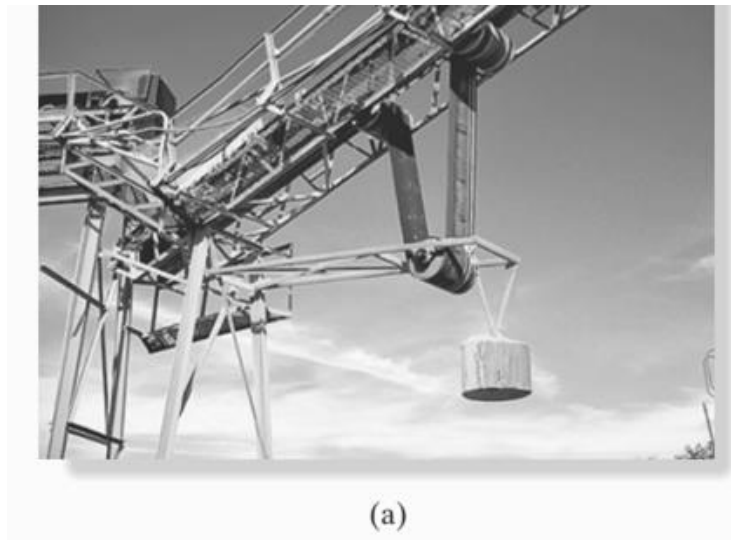
Frames and Machines: Free-Body Diagrams

Frame example:



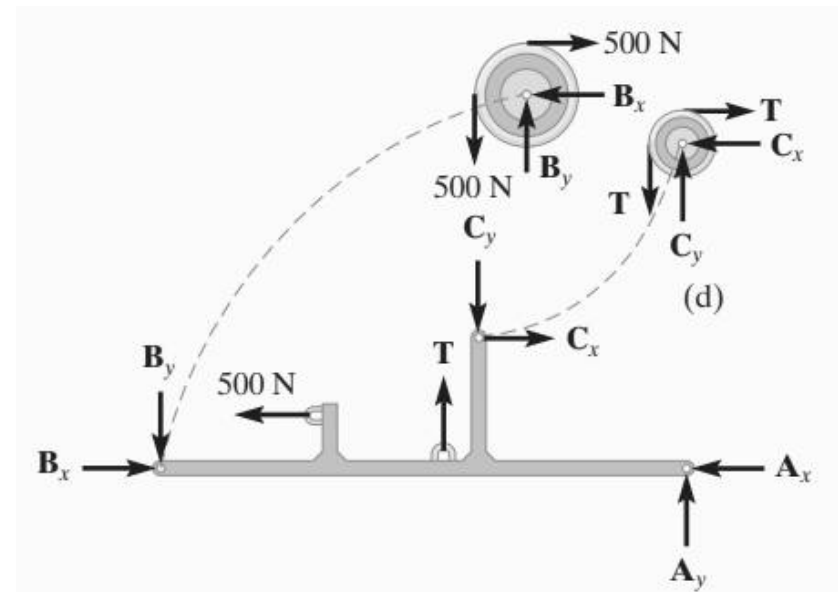
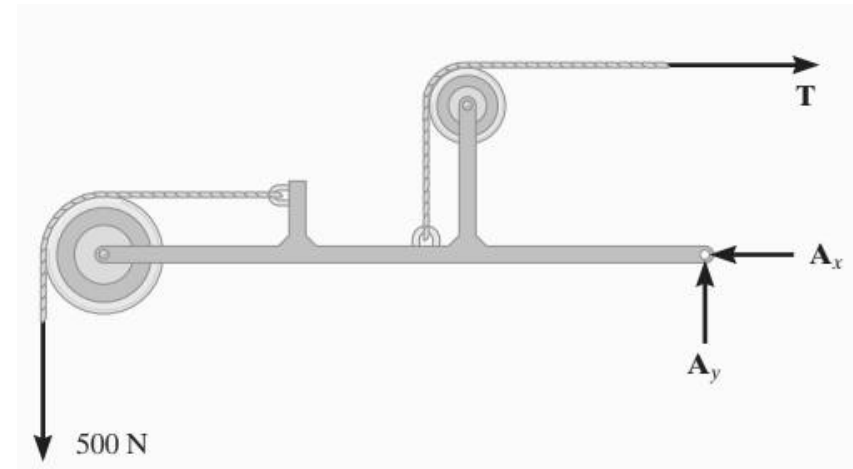
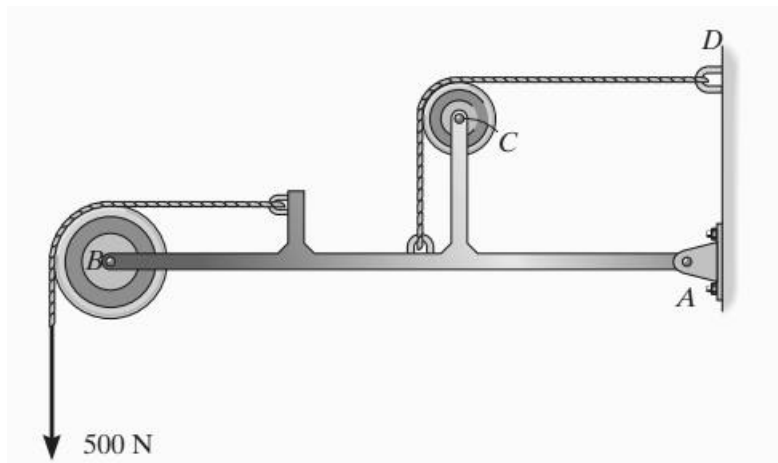
Frames and Machines: Free-Body Diagrams

Conveyor belt example:



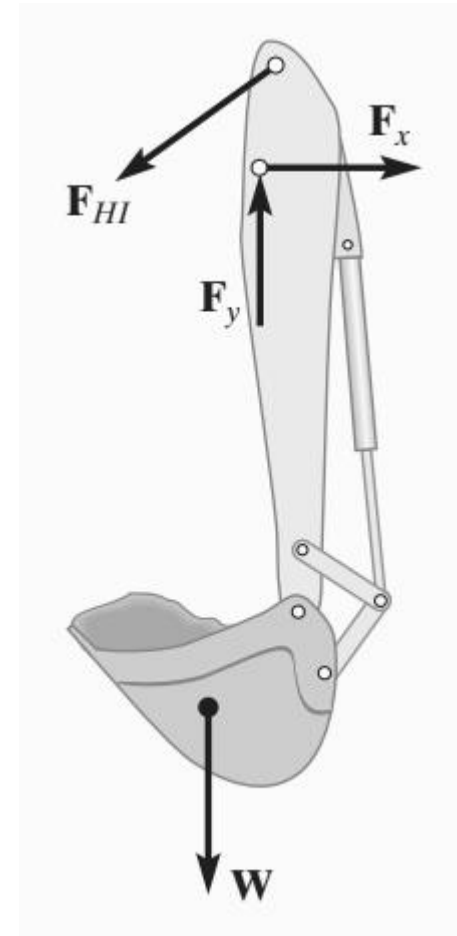
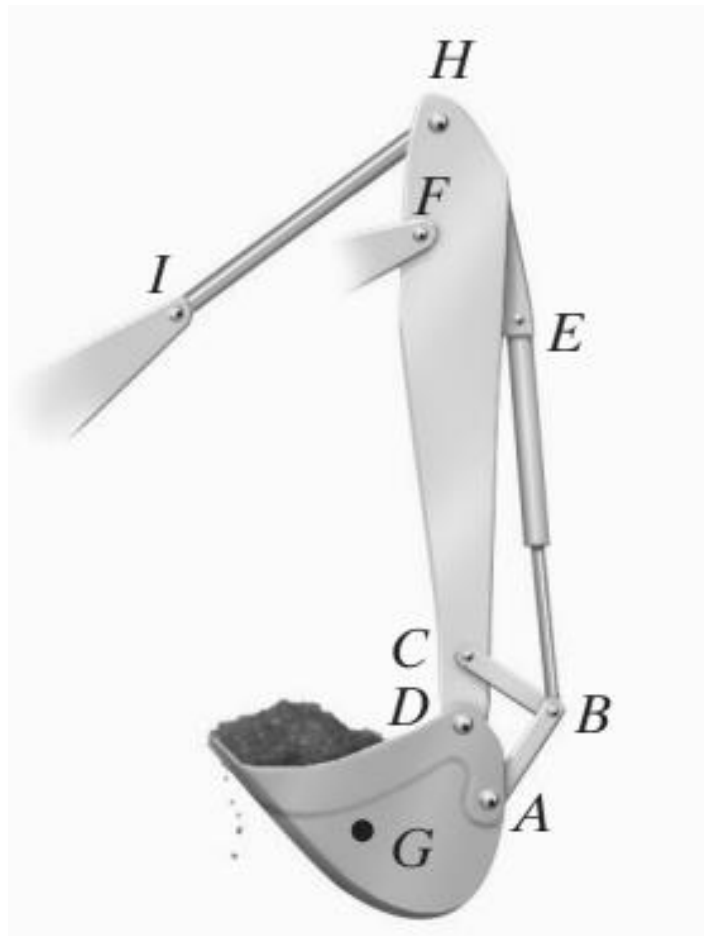
Frames and Machines: Free-Body Diagrams

Frame example:



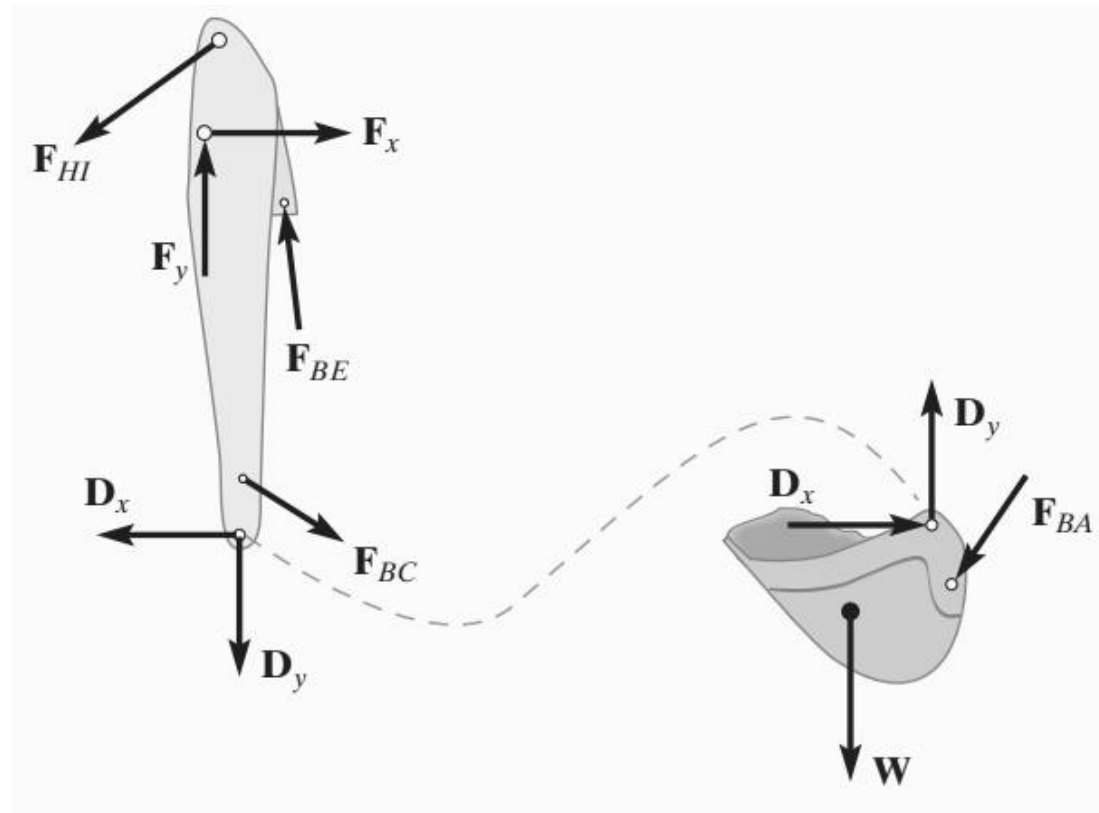
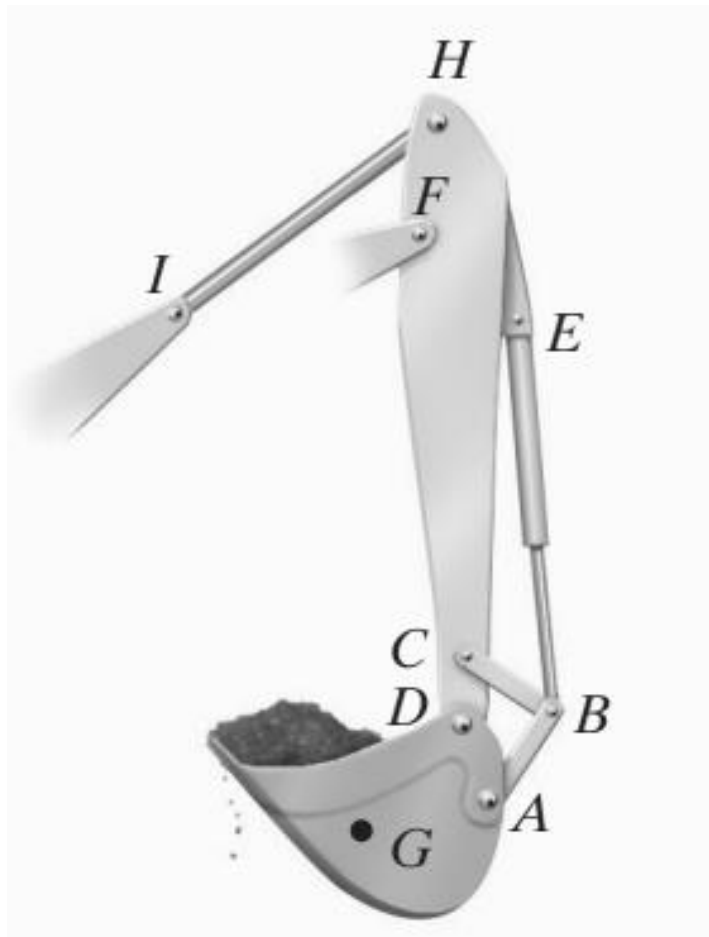
Frames and Machines: Free-Body Diagrams

Backhoe example:



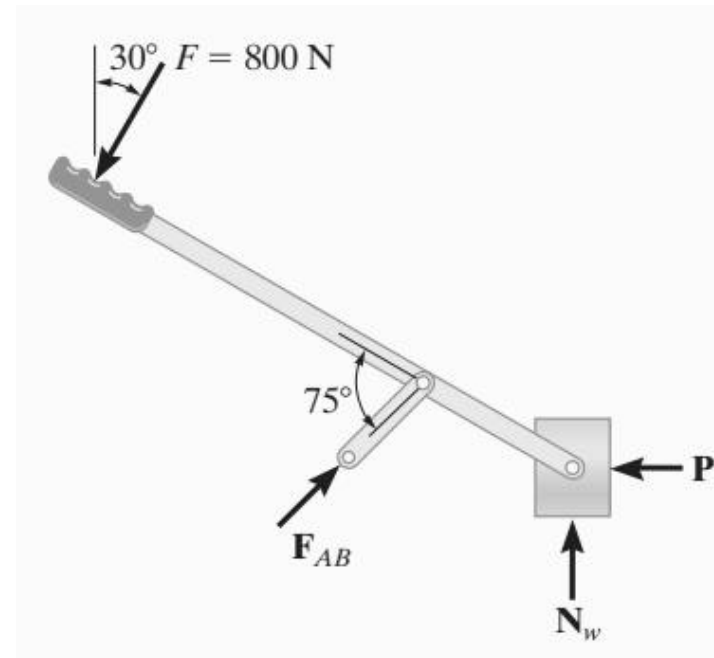
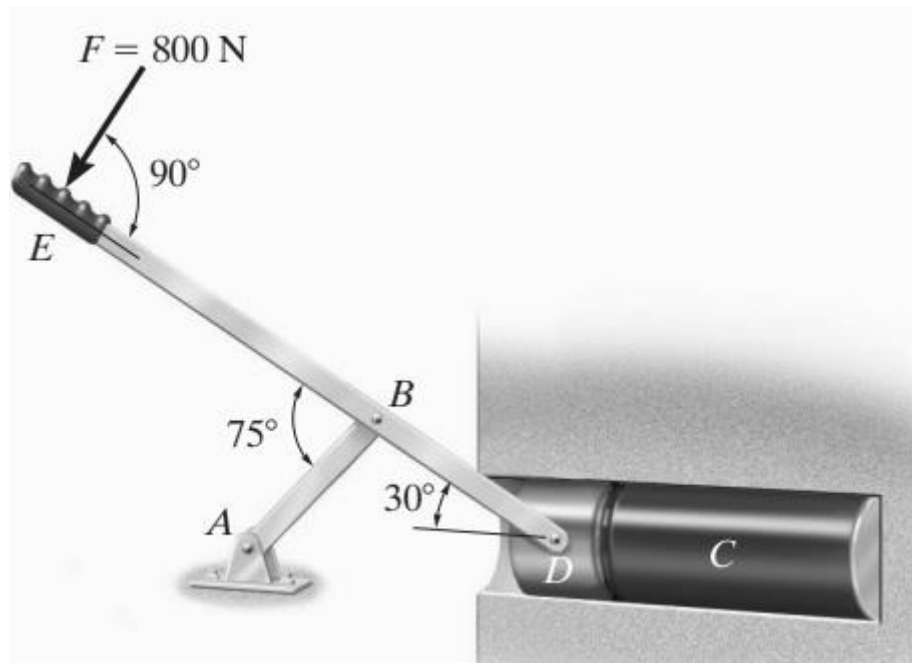
Frames and Machines: Free-Body Diagrams

Backhoe example:



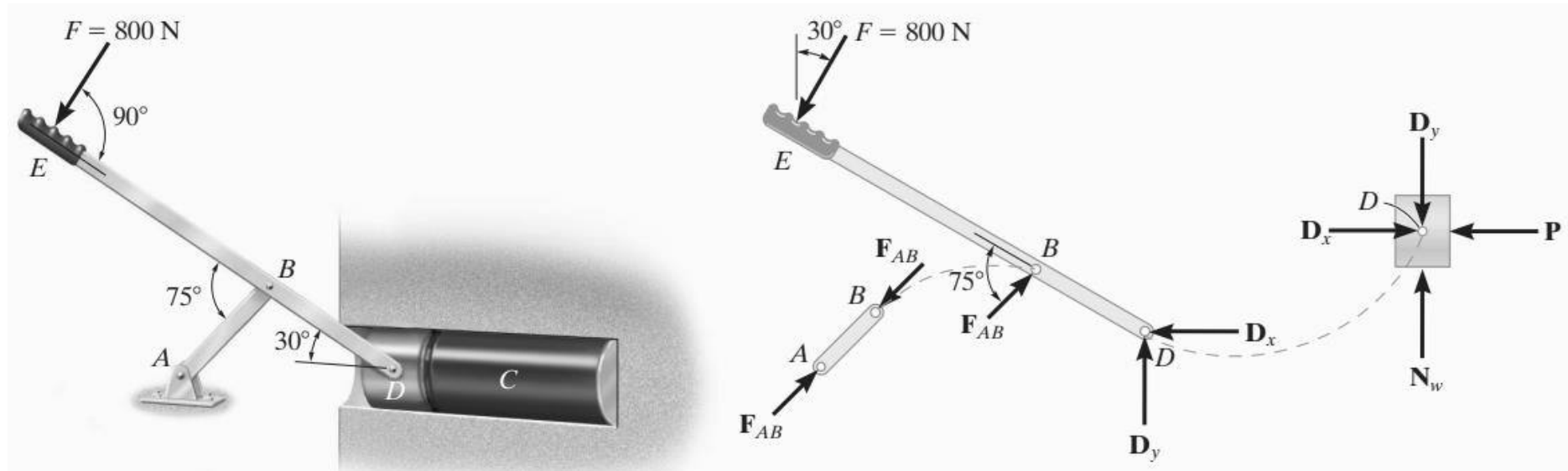
Frames and Machines: Free-Body Diagrams

Smooth piston and link mechanism example:

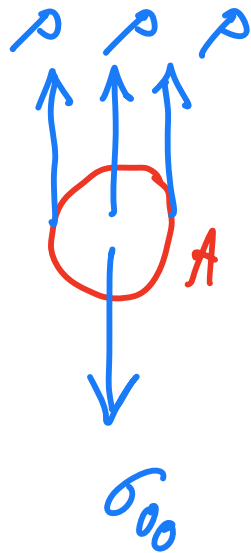


Frames and Machines: Free-Body Diagrams

Smooth piston and link mechanism example:

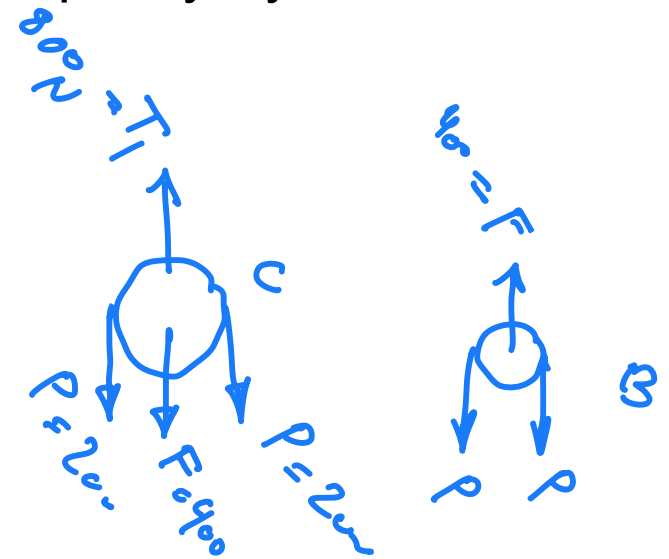
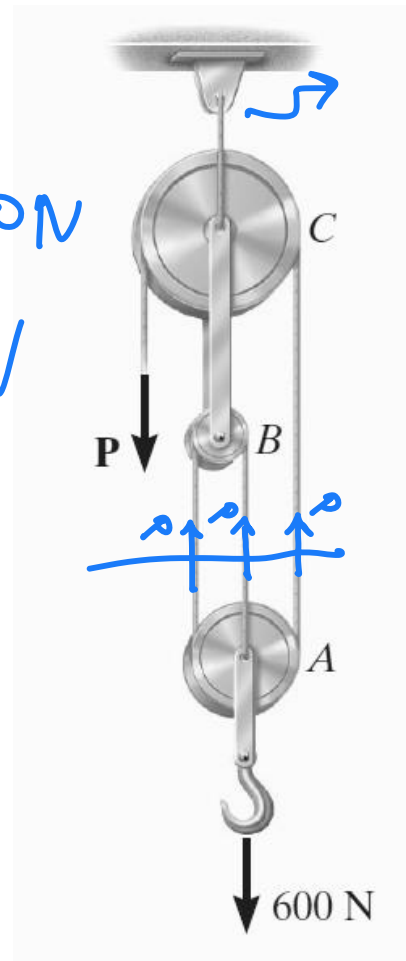


Example: Determine the tension in the cables and the force P required to support the 600-N force using the frictionless pulley system shown below.

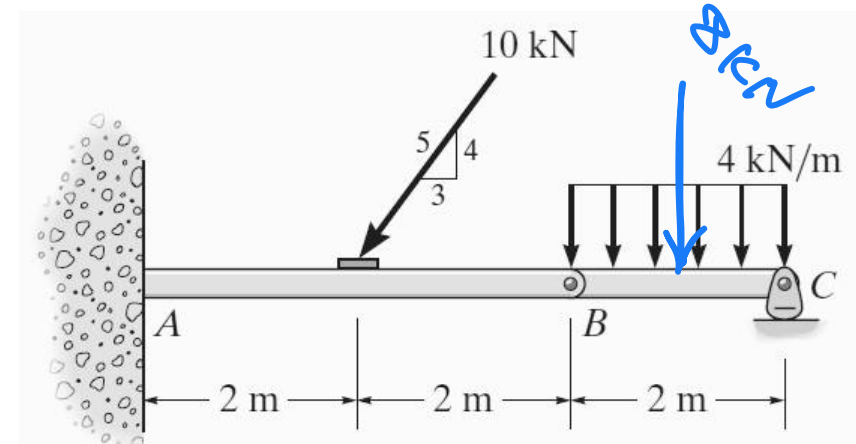
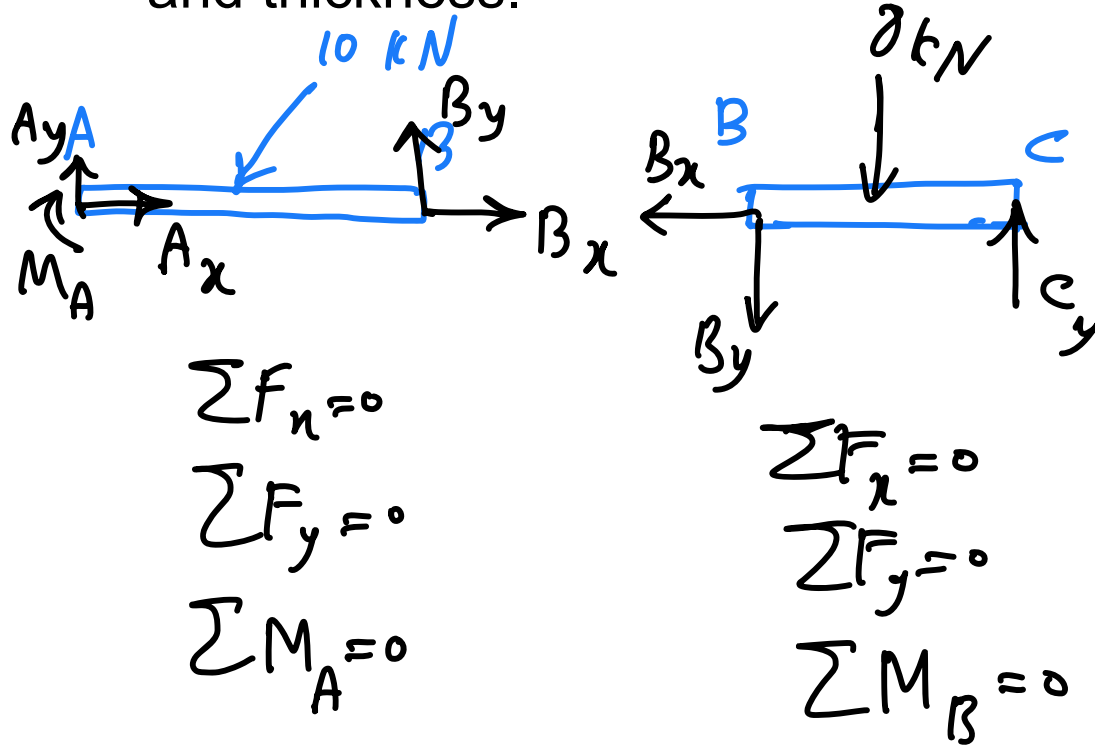


$$3p = 600 \text{ N}$$

$$p = 200 \text{ N}$$



Example: The compound beam shown below is pin connected at B . Determine the components of reaction at its supports. Neglect its weight and thickness.

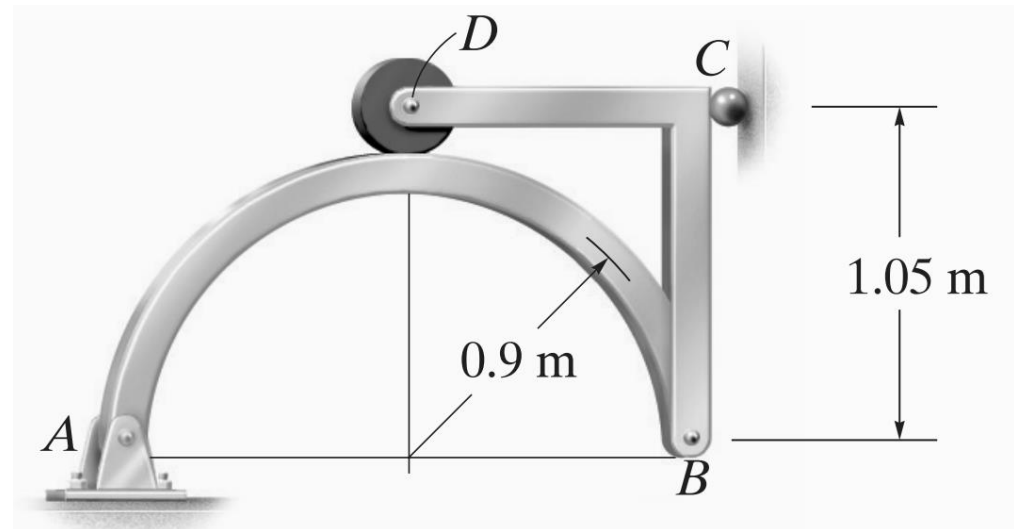
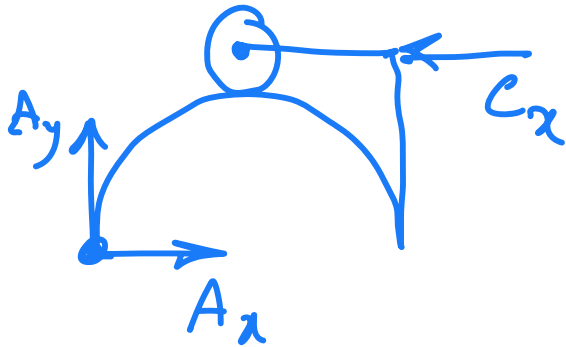


$$F = \int w dx = A$$

$$\int x w dx = \bar{x} F$$

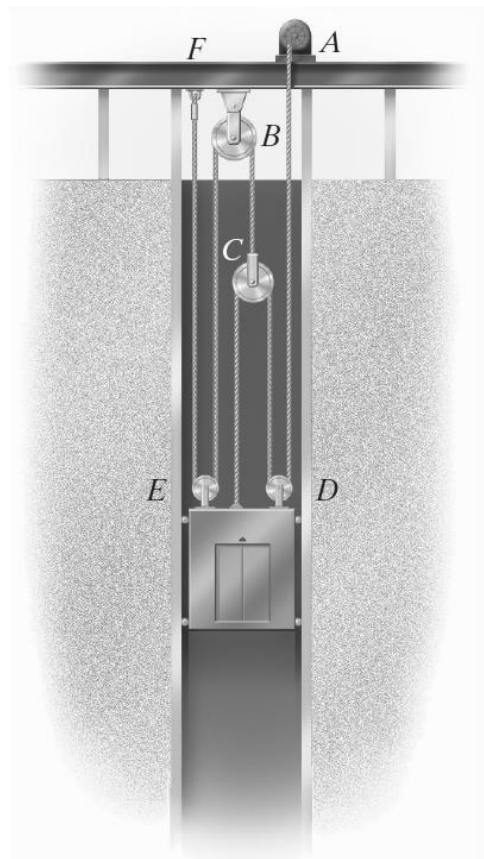
$$\bar{x} = \frac{\int x w dx}{F} = \frac{\int x w dx}{\int w dx}$$

Example: The smooth disk shown below is pinned at D and has a weight of 90 N. Neglecting the weights of the other members, determine the horizontal and vertical components of reaction at pins B and D .



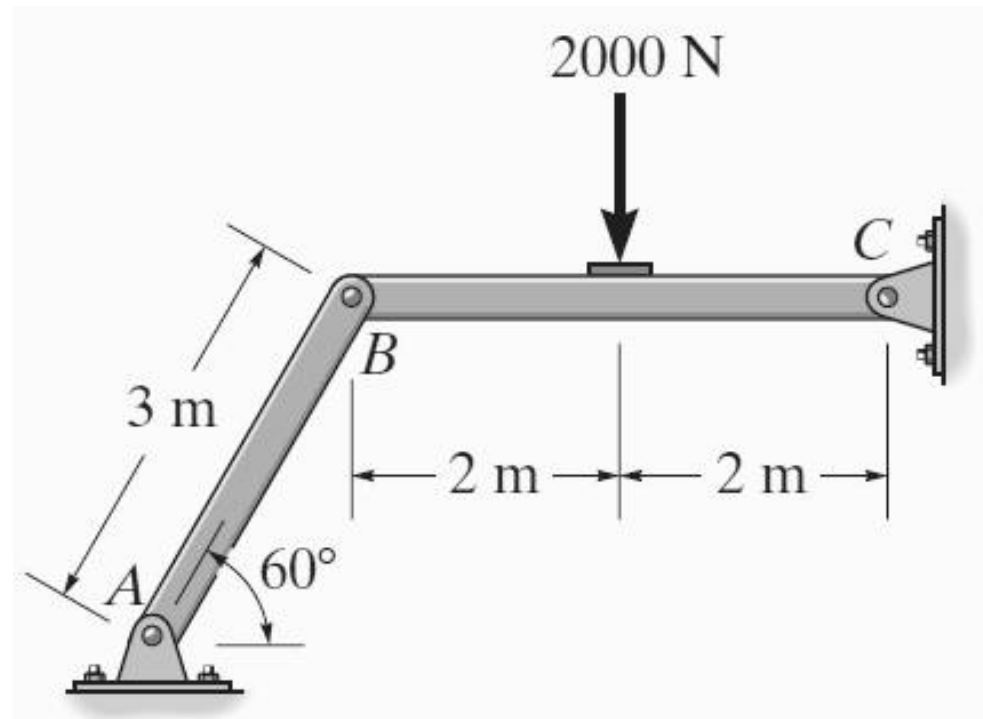
Practice at Home

1. The 500-kg elevator car shown below is being hoisted at constant speed by motor A using the pulley system shown. Determine the force developed in the two cables.



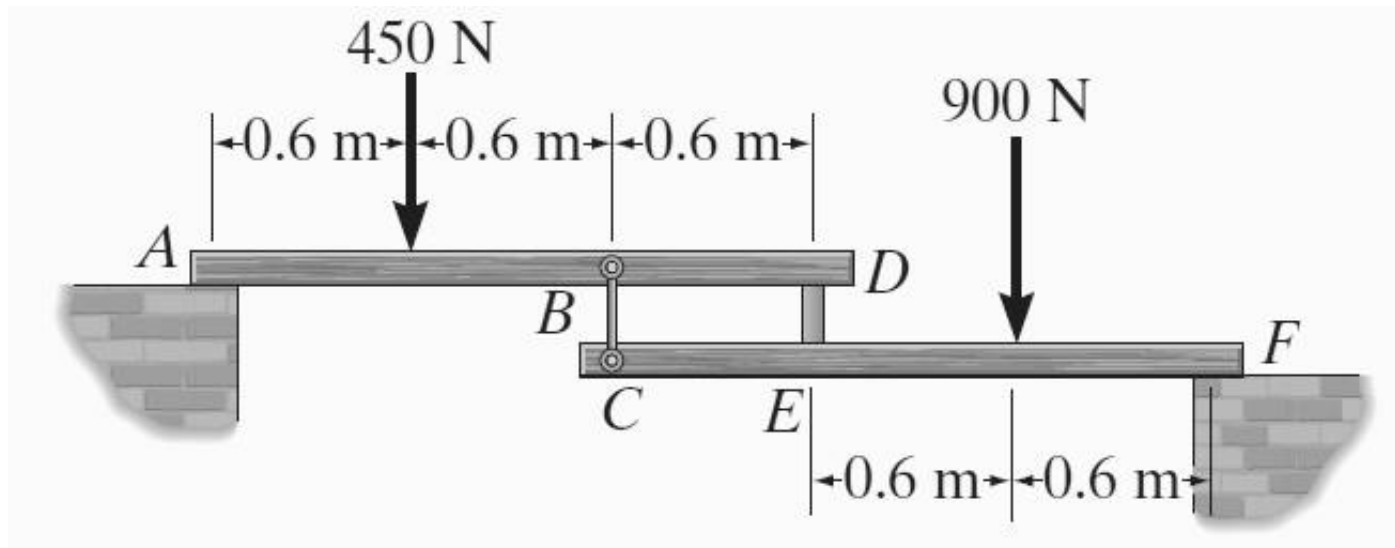
Practice at Home

2. Determine the horizontal and vertical components of force which the pin at C exerts on member BC of the frame shown below.



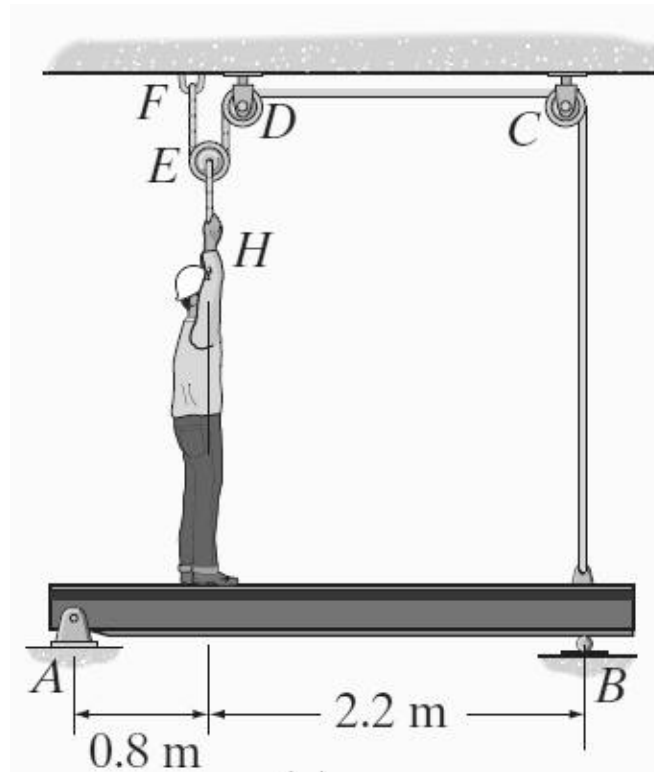
Practice at Home

3. The two planks shown below are connected by cable BC and a smooth spacer DE . Determine the reactions at the smooth supports A and F , and find the force developed in the cable and spacer.



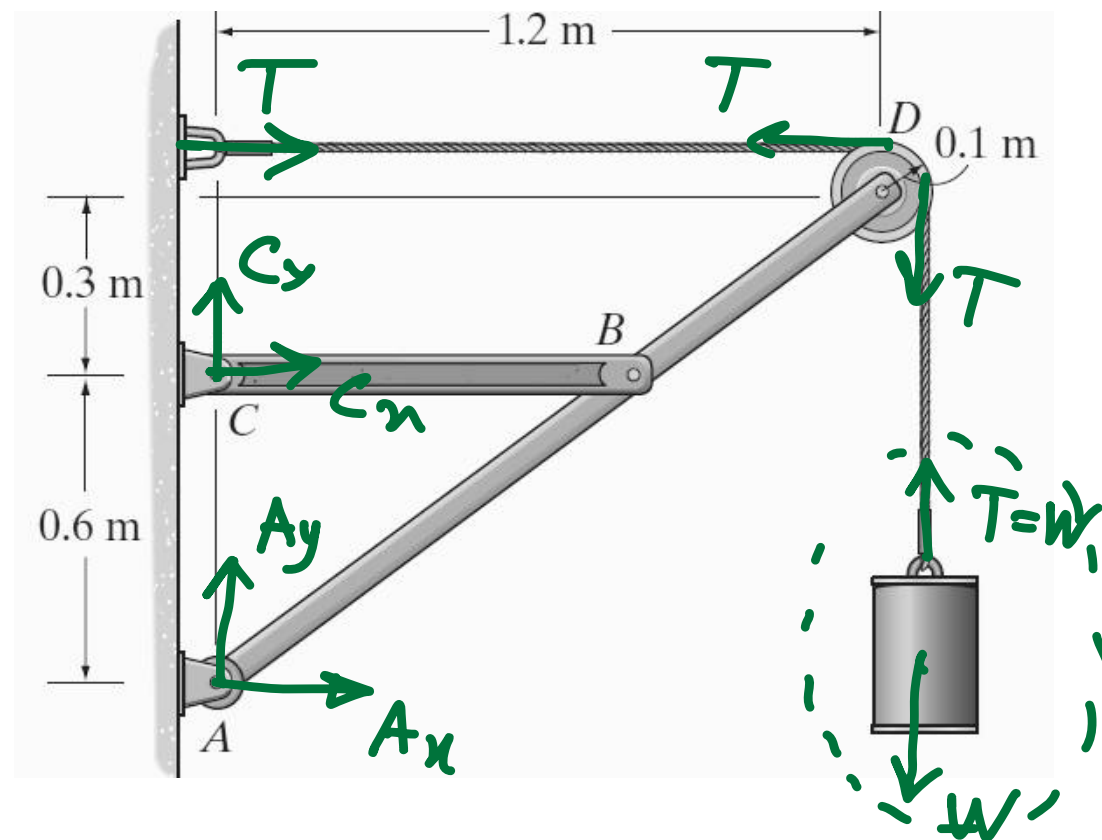
Practice at Home

4. The 75-kg man shown below attempts to lift the 40-kg uniform beam off the roller support at B . Determine the tension developed in the cable attached to B and the normal reaction of the man on the beam when this is about to occur.



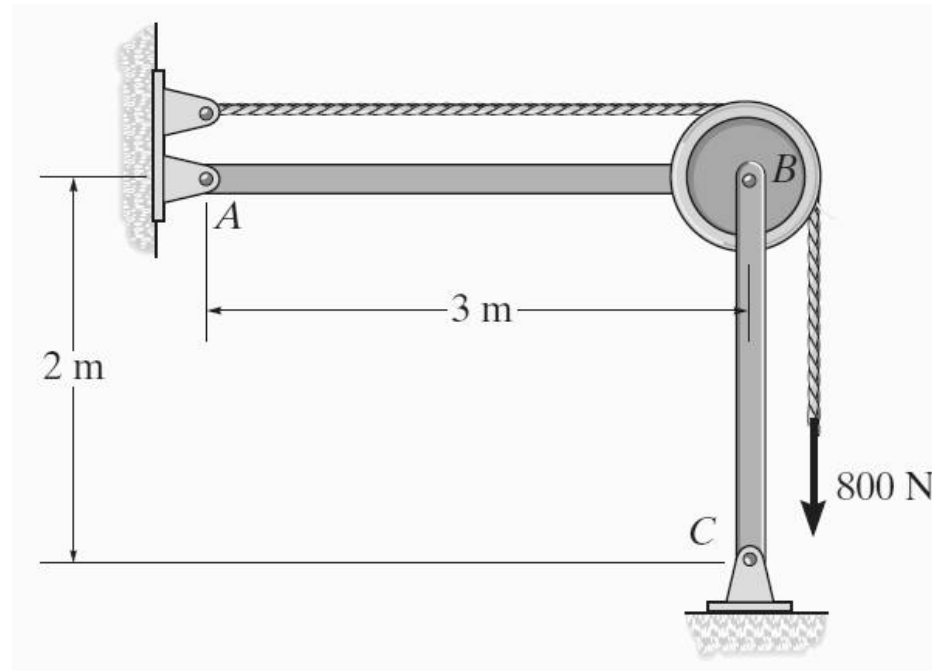
Practice at Home

5. The frame below supports the 50-kg cylinder. Determine the horizontal and vertical components of reaction at A and the force at C.



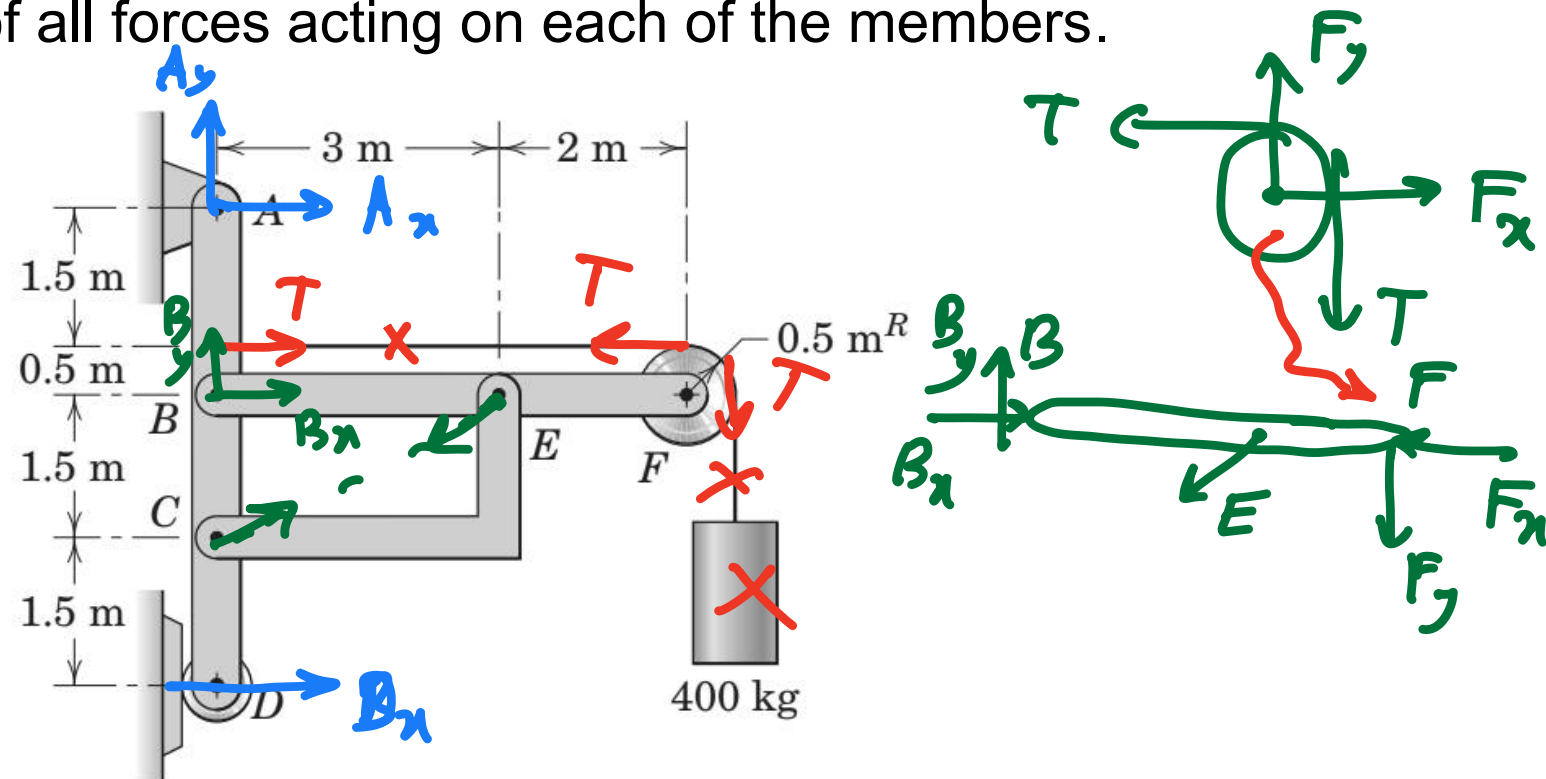
Practice at Home

6. Determine the force the pins at A and B exert on the two-member frame shown below.

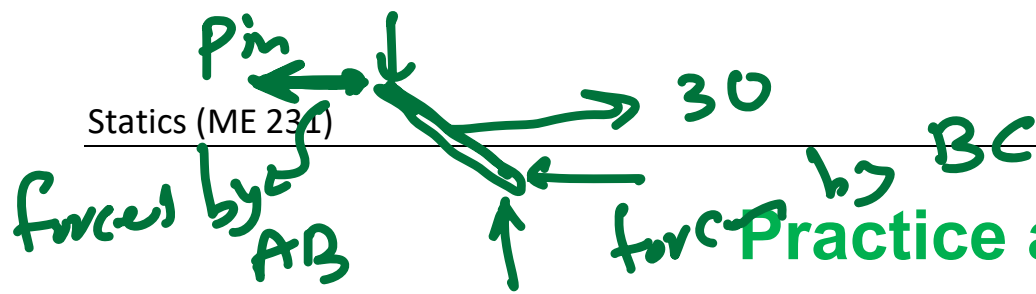


Practice at Home

7. The frame supports the 400-kg load in the manner shown. Neglect the weights of the members compared with the forces induced by the load and compute the horizontal and vertical components of all forces acting on each of the members.

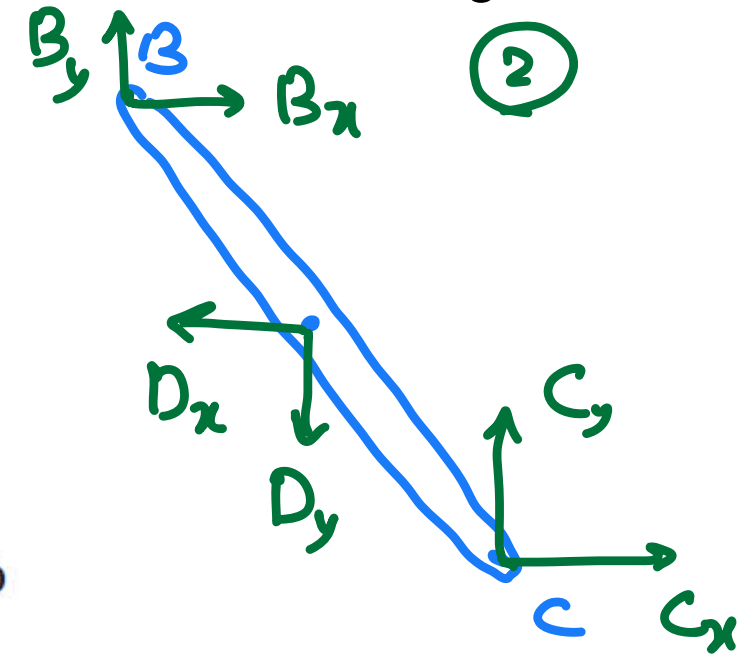
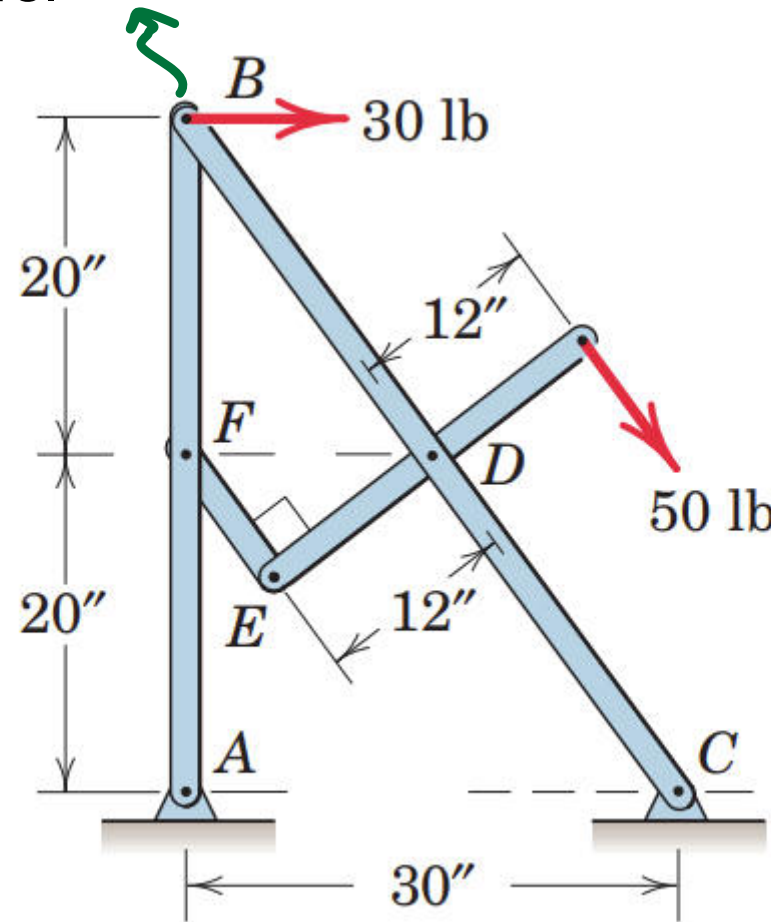
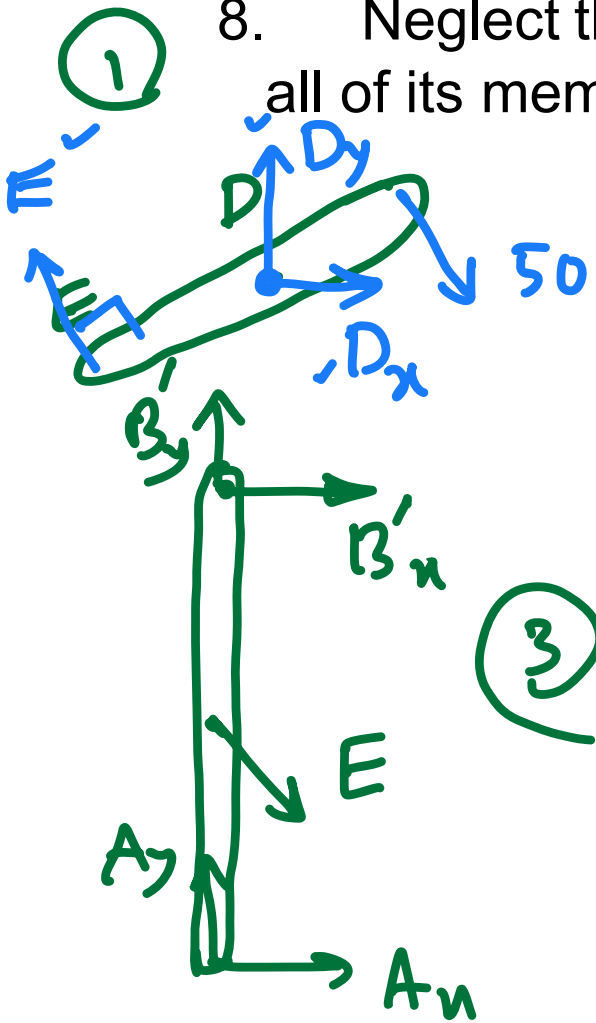


$$B'_y = B_y, \quad B_x + B'_x = 30$$



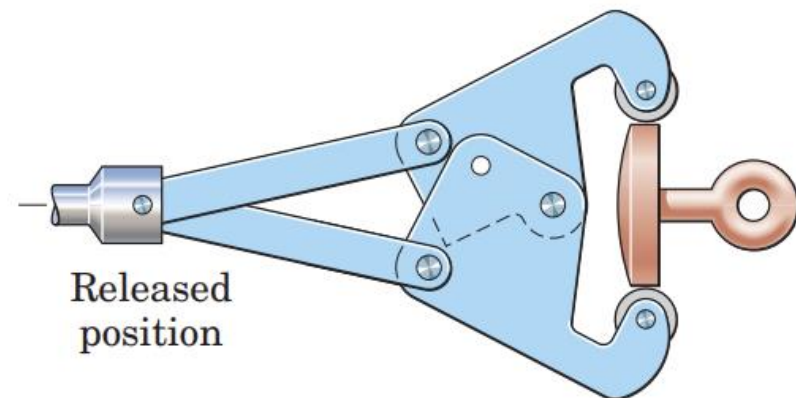
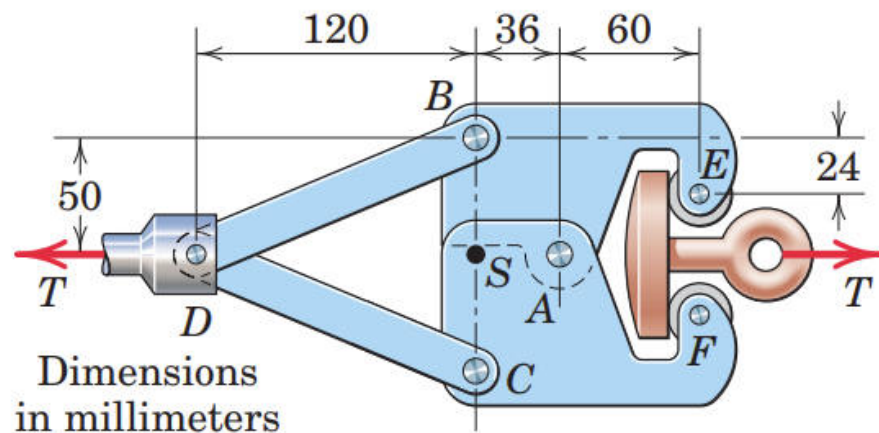
Practice at Home

8. Neglect the weight of the frame and compute the forces acting on all of its members.



Practice at Home

9. The machine shown is designed as an overload protection device which releases the load when it exceeds a predetermined value T . A soft metal shear pin S is inserted in a hole in the lower half and is acted on by the upper half. When the total force on the pin exceeds its strength, it will break. The two halves then rotate about A under the action of the tensions in BD and CD , as shown in the second sketch, and rollers E and F release the eye bolt. Determine the maximum allowable tension T if the pin S will shear when the total force on it is 800 N. Also compute the corresponding force on the hinge pin A .



Practice at Home

10. In the position shown, the excavator applies a 20-kN force parallel to the ground. There are two hydraulic cylinders AC to control the arm OAB and a single cylinder DE to control arm $EBIF$. (a) Determine the force in the hydraulic cylinders AC and the pressure p_{AC} against their pistons, which have an effective diameter of 95 mm. (b) Also determine the force in hydraulic cylinder DE and the pressure p_{DE} against its 105-mm-diameter piston. Neglect the weights of the members compared with the effects of the 20-kN force.

